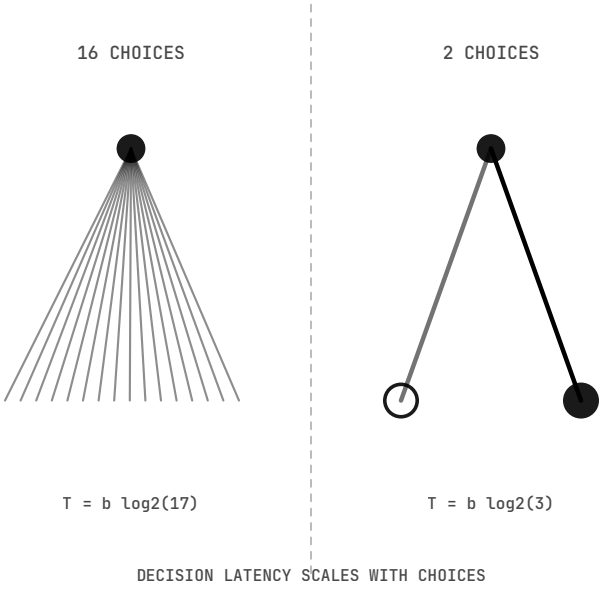
Hick's Law

Decision time increases logarithmically with the number and informational complexity of available choices.



WHEN TO APPLY

Users stall or scan repeatedly when a menu or settings page presents many flat, unranked options.

DEC-01

Hick's Law

OPERATIONAL CHECKLIST

- [ ] Cap flat options in top-level navigation; each added option lengthens the decision.
- [ ] Group large select menus into categories or a filterable combobox.
- [ ] Pre-highlight a recommended default to skip part of the decision tree.

ANTI-PATTERN VS REMEDIATION

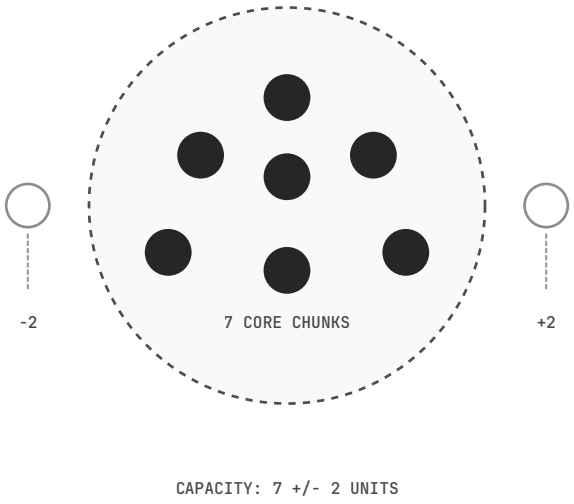
Violation: An enterprise settings page offers a flat, unordered dropdown of 85 country options.

Fix: Replace it with an autocomplete combobox: locale auto-detect, keyboard filtering, common options first.

Source: William Edmund Hick (1952). *On the Rate of Gain of Information.* Quarterly Journal of Experimental Psychology, 4(1), 11-26.

Miller's Law

Working memory capacity spans approximately seven, plus or minus two, discrete information chunks.



CAPACITY: 7 +/- 2 UNITS

WHEN TO APPLY

Users lose track of values across long unbroken strings, forms, or dashboards with many competing figures.

DEC-02

Miller's Law

OPERATIONAL CHECKLIST

- [ ] Chunk long strings: format card numbers as four groups of four digits.
- [ ] Split multi-step checkout into three to five navigable phases.
- [ ] Cap visible summary cards on an overview dashboard at seven key indicators.

ANTI-PATTERN VS REMEDIATION

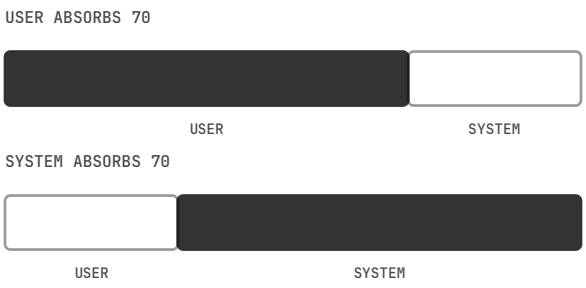
Violation: A payment field demands a raw 16-digit card number in one unformatted text string.

Fix: Parse input into visible groups as typed (4111 0000 0000 1111) so it can be verified quickly.

Source: George A. Miller (1956). *The Magical Number Seven, Plus or Minus Two: Some Limits on Our Capacity for Processing Information.* Psychological Review, 63(2), 81-97.

Tesler's Law

Every system carries irreducible complexity; design moves it between the system and the user.



COMPLEXITY MOVES; THE TOTAL DOES NOT

WHEN TO APPLY

People are asked to supply details the product could derive, such as time zones, currency formats, or bank codes.

DEC-03

Tesler's Law

OPERATIONAL CHECKLIST

- [ ] Derive time zones, currency formats, and country codes from context instead of prompting.
- [ ] Resolve file type, encoding, and MIME parameters during upload rather than asking.
- [ ] Ship production-ready defaults for advanced settings so people opt out, not in.

ANTI-PATTERN VS REMEDIATION

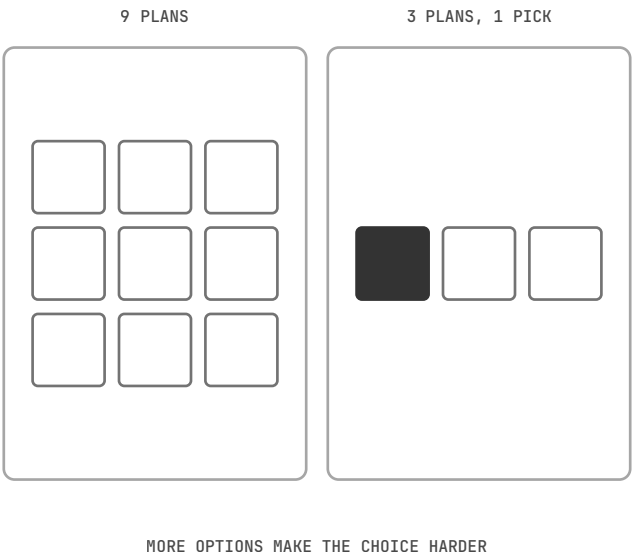
Violation: An international payment form asks customers for SWIFT, routing, clearing, and branch details by hand.

Fix: Accept the IBAN and resolve the bank name, branch, and routing codes in the backend.

Source: Dan Saffer (2007). *Designing for Interaction.* New Riders, first edition.

Choice Overload

Too many comparable options stall decisions and leave people less satisfied with what they choose.



WHEN TO APPLY

A pricing page or settings screen shows many similar options and people stall, default, or leave without choosing.

DEC-04

Choice Overload

OPERATIONAL CHECKLIST

- [ ] Limit consumer pricing to three or four plans with visibly different value.
- [ ] Mark one option as recommended so a default path through the comparison is obvious.
- [ ] Move advanced filters and rare configurations behind progressive disclosure.

ANTI-PATTERN VS REMEDIATION

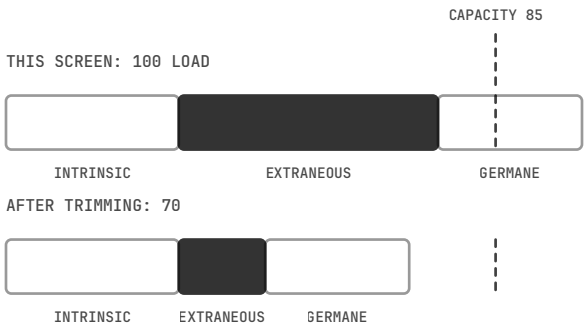
Violation: A pricing table shows nine overlapping plans with no recommended option, and visitors leave without choosing.

Fix: Offer three clear tiers, mark one recommended, and route custom plans to a separate inquiry.

Source: Sheena S. Iyengar & Mark R. Lepper (2000). *When Choice Is Demotivating: Can One Desire Too Much of a Good Thing?* Journal of Personality and Social Psychology, 79(6), 995-1006.

Cognitive Load Theory

Interaction suffers when extraneous cognitive demands consume the working memory a task needs.



LOAD PAST CAPACITY COSTS THE TASK

WHEN TO APPLY

People re-read instructions, hunt for guidance, or make avoidable errors in a dense single-page form.

DEC-05

Cognitive Load Theory

OPERATIONAL CHECKLIST

- [ ] Remove decoration, redundant banners, and instructions that repeat the visible label.
- [ ] Reveal secondary detail only when the current step needs it.
- [ ] Put guidance beside the field it explains instead of on a separate help page.

ANTI-PATTERN VS REMEDIATION

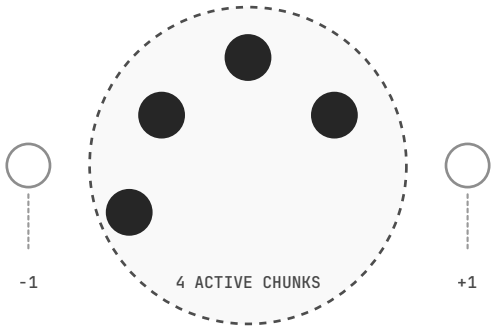
Violation: A tax form presents 100 fields on one scrolling page, each preceded by paragraphs of legal text.

Fix: Split filing into focused steps and move legal detail into inline popovers.

Source: John Sweller (1988). *Cognitive Load During Problem Solving: Effects on Learning.* Cognitive Science, 12(2), 257-285.

Working Memory Capacity Model

Active memory holds about four chunks, so take the task's burden out of memory.



CAPACITY: 4 +/- 1 CHUNKS

WHEN TO APPLY

A multi-step flow asks people to remember a value from an earlier step while entering data on a later one.

DEC-06

Working Memory Capacity Model

OPERATIONAL CHECKLIST

- [ ] Keep earlier answers visible while someone enters related data in a later step.
- [ ] Pin summary headers and context panels while people scroll a long table.
- [ ] Show breadcrumbs and a persistent cart through checkout.
- [ ] Put a copy button beside any value people must reuse on another screen.

ANTI-PATTERN VS REMEDIATION

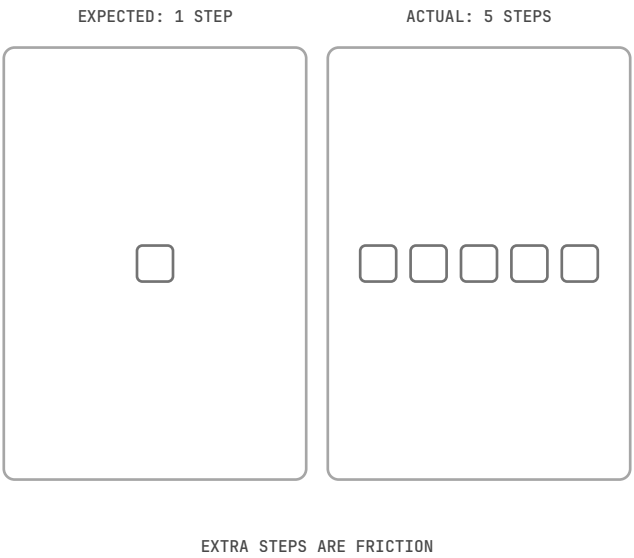
Violation: A four-step wizard makes people memorize a key shown in step one and retype it in step four.

Fix: Step four shows the key in a read-only card with a copy button.

Source: Nelson Cowan (2001). *The Magical Number 4 in Short-Term Memory: A Reconsideration of Mental Storage Capacity.* Behavioral and Brain Sciences, 24(1), 87-114.

Cognitive Friction

Interfaces create friction when reaching a goal takes more or different actions than people expect.



WHEN TO APPLY

Users click a search icon, wait for the overlay, then click the field again before they can type.

DEC-07

Cognitive Friction

OPERATIONAL CHECKLIST

- [ ] Let Enter submit single-field forms such as search and sign-in.
- [ ] Move focus into the primary input as soon as a dialog opens.
- [ ] Keep Tab, Shift+Tab, and arrow keys working through every menu.

ANTI-PATTERN VS REMEDIATION

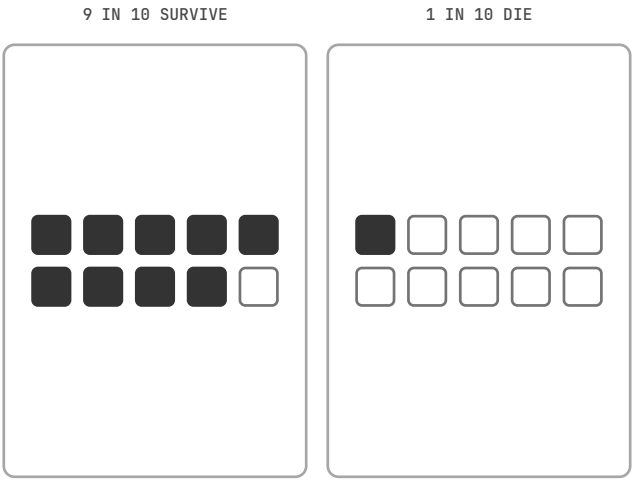
Violation: Opening search means clicking the icon, waiting for the overlay, and clicking the field before typing.

Fix: Bind search to a keyboard shortcut that opens the dialog already focused on its input.

Source: Alan Cooper (1999). *The Inmates Are Running the Asylum.* Sams Publishing.

Framing Effect

People decide differently when the same facts are described as gains rather than losses.



SAME TEN CASES, TWO DESCRIPTIONS

WHEN TO APPLY

Two screens state one outcome, but one stresses what is gained and the other what is lost, and choices diverge.

DEC-08

Framing Effect

OPERATIONAL CHECKLIST

- [ ] State availability as uptime gained (99.9%) rather than downtime risk.
- [ ] Present discounts as money saved rather than total billed.
- [ ] Describe security work as protecting data rather than as a failure warning.

ANTI-PATTERN VS REMEDIATION

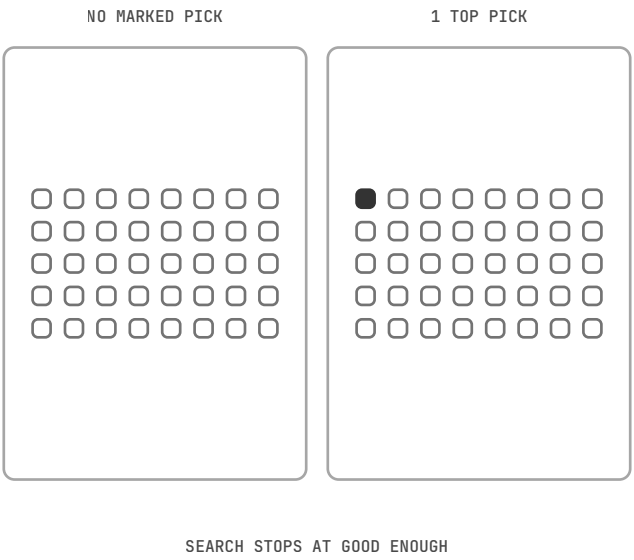
Violation: A backup prompt warns that refusing leaves all records open to total loss.

Fix: The same prompt explains that enabling backup protects local files against hardware failure.

Source: Amos Tversky & Daniel Kahneman (1981). *The Framing of Decisions and the Psychology of Choice.* Science, 211(4481), 453-458.

Simon's Bounded Rationality and Satisficing

People accept the first option that clears their threshold rather than searching for the best.



WHEN TO APPLY

People page through all forty search results because nothing marks which listing is good enough to take.

DEC-09

Simon's Bounded Rationality and Satisficing

OPERATIONAL CHECKLIST

- [ ] Surface a clearly marked top pick with its rating or badge above the result list.
- [ ] Show the criteria that make an option acceptable beside it, not one detail page away.
- [ ] Keep essential filters visible instead of requiring a boolean query to narrow the set.

ANTI-PATTERN VS REMEDIATION

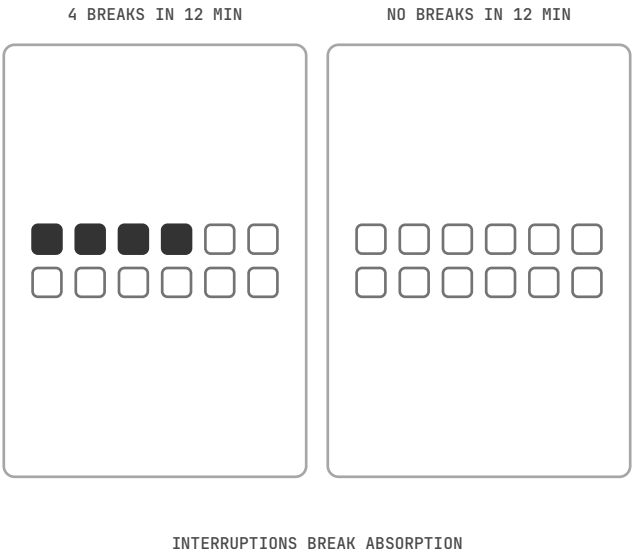
Violation: A hotel search makes people open all forty listings because none shows rating, cancellation terms, and price together.

Fix: A Top Pick card above the grid shows rating, cancellation terms, and verified price for those dates.

Source: Herbert A. Simon (1956). *Rational Choice and the Structure of the Environment.* Psychological Review, 63(2), 129-138.

Cognitive Absorption

Absorbed people lose track of time and stop noticing the interface around the task.



WHEN TO APPLY

A notification, promotional banner, or modal interrupts a writing or coding session that was running smoothly.

DEC-10

Cognitive Absorption

OPERATIONAL CHECKLIST

- [ ] Silence non-urgent notifications, promotional banners, and modal interruptions during focused work.
- [ ] Save in the background and report status without a dialog or a blocking spinner.
- [ ] Offer a distraction-free mode and keyboard paths for sustained work.

ANTI-PATTERN VS REMEDIATION

Violation: A code editor shows rating prompts and marketing toasts every ten minutes while someone is writing.

Fix: A focus mode hides ambient chrome, holds notifications, autosaves silently, and shows status in a quiet corner.

Source: Ritu Agarwal & Elena Karahanna (2000). *Time Flies When You're Having Fun: Cognitive Absorption and Beliefs about Information Technology Usage.* MIS Quarterly, 24(4), 665-694.

Decision Fatigue

Decision quality degrades over a long session of repetitive choices.



WHEN TO APPLY

A flow reaches its twelfth configuration question, and the last three answers were left at the default.

DEC-11

Decision Fatigue

OPERATIONAL CHECKLIST

- [ ] Batch small choices into one review step instead of asking them one at a time.
- [ ] Defer non-critical configuration until the main task is finished.
- [ ] Pre-populate safe defaults so most questions need no answer.

ANTI-PATTERN VS REMEDIATION

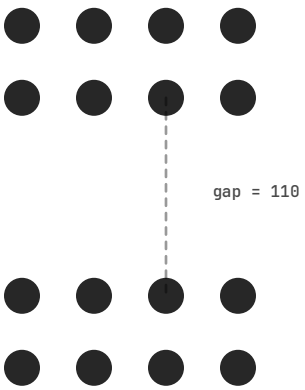
Violation: An onboarding flow asks twenty separate configuration questions before the first task, each with no default.

Fix: One review screen batches the questions, defers the optional half, and pre-fills safe defaults.

Source: Roy F. Baumeister, Ellen Bratslavsky, Mark Muraven & Dianne M. Tice (1998). *Ego Depletion: Is the Active Self a Limited Resource?* Journal of Personality and Social Psychology, 74(5), 1252-1265.

Law of Proximity

Objects positioned close to one another are perceived as a unified conceptual group.



SPACING CREATES THE GROUPS

WHEN TO APPLY

Form labels look attached to the wrong input, or unrelated actions read as one group because spacing is uniform.

GES-01

Law of Proximity

OPERATIONAL CHECKLIST

- [ ] Place labels closer to their own input than to the preceding field (4px vs 24px).
- [ ] Keep spacing inside a card tighter (8-12px) than spacing between cards (24-32px).
- [ ] Group a primary action with its explanatory copy inside one layout cluster.

ANTI-PATTERN VS REMEDIATION

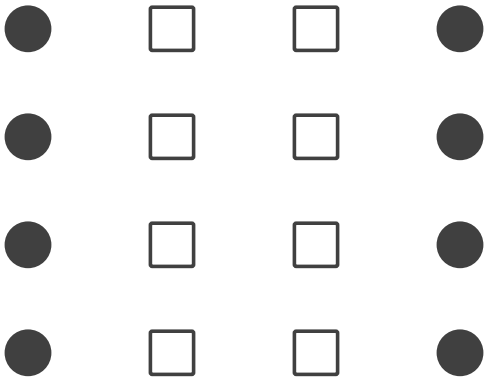
Violation: A form uses equal 16px margins everywhere, so each input visually attaches to the label below it.

Fix: Use 4px from label to input and 24px from the input container to the next field label.

Source: Max Wertheimer (1923). *Untersuchungen zur Lehre von der Gestalt II.* Psychologische Forschung, 4, 301-350.

Law of Similarity

Elements that share colour, shape, or size are read as one kind of thing.



ALIKE BY FORM, NOT BY PLACE

WHEN TO APPLY

Two different kinds of control share a style, or one kind is drawn three different ways.

GES-02

Law of Similarity

OPERATIONAL CHECKLIST

- [ ] Give every primary action the same fill, radius, and weight; a variant is a different thing.
- [ ] Draw metadata badges and filter pills in different shapes, not just different colours.
- [ ] Reserve one accent colour for items that must stand apart from everything else.
- [ ] Squint at a screen: if unlike things merge into blobs, the shared style is the defect.

ANTI-PATTERN VS REMEDIATION

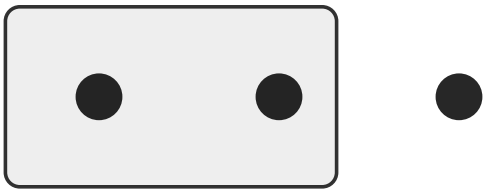
Violation: Submit buttons appear as blue pills, green rectangles, and outlines on different screens.

Fix: All primary actions use one token: same background, text colour, radius, and weight.

Source: Max Wertheimer (1923). *Untersuchungen zur Lehre von der Gestalt II.* Psychologische Forschung, 4, 301-350.

Law of Common Region

A boundary drawn around things makes them one group, even when they are far apart.



ONE LINE DRAWS THE GROUP

WHEN TO APPLY

A dashboard has no borders, so metrics read as one undifferentiated field.

GES-03

Law of Common Region

OPERATIONAL CHECKLIST

- [ ] Put related fields inside one card surface with a single border or fill.
- [ ] Keep at least 16px of padding inside a region so its contents read as contained.
- [ ] Do not wrap every row in its own border; use a boundary only where grouping matters.
- [ ] Test whether removing the boundary changes the grouping users report; if not, remove it.

ANTI-PATTERN VS REMEDIATION

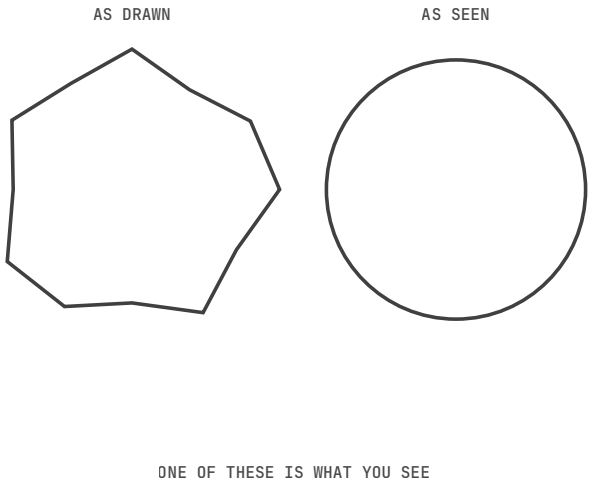
Violation: Thirty metrics sit on a bare canvas with only whitespace between them, so no one can tell which belong together.

Fix: Metrics sit in bordered cards with a faint background, each card holding one functional domain.

Source: Stephen E. Palmer (1992). *Common region: A new principle of perceptual grouping*. Cognitive Psychology, 24(3), 436-447.

Law of Prägnanz (Law of Simplicity / Good Figure)

Of the readings a shape allows, perception takes the simplest and most regular.



WHEN TO APPLY

A layout can be parsed two ways, and the harder reading is the one users settle on.

GES-04

Law of Prägnanz (Law of Simplicity / Good Figure)

OPERATIONAL CHECKLIST

- [ ] Align cards and panels to one grid so their edges resolve into a few straight lines.
- [ ] Strip icon detail until the silhouette reads at a glance; keep one recognisable outline.
- [ ] Replace diagonal cuts and stepped dividers with rectangles on the same axes.
- [ ] Show a layout to someone for two seconds and ask what the page is made of.

ANTI-PATTERN VS REMEDIATION

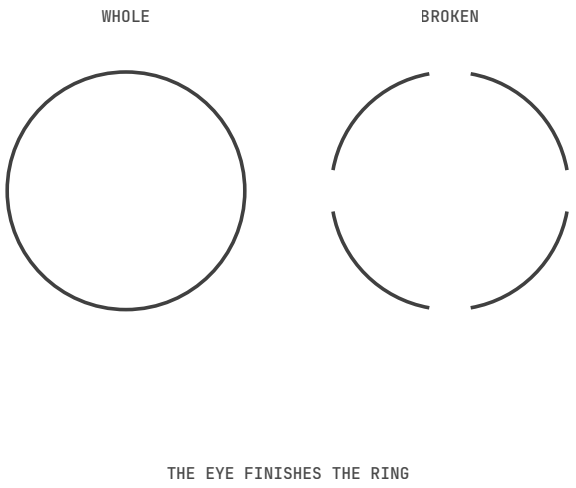
Violation: Cards are skewed and cut on diagonals with stepped dividers, so every block needs tracing.

Fix: Cards are rectangles on one grid, with edges that line up and a readable silhouette.

Source: Max Wertheimer (1923). *Untersuchungen zur Lehre von der Gestalt II.* Psychologische Forschung, 4, 301-350.

Law of Closure

The eye completes a broken outline and sees the whole shape it implies.



WHEN TO APPLY

A carousel stops exactly at the screen edge, so nothing signals more content.

GES-05

Law of Closure

OPERATIONAL CHECKLIST

- [ ] Let the next card peek past the edge instead of ending the list flush with the viewport.
- [ ] Draw icons as open strokes where the missing segment does not change the meaning.
- [ ] Use skeleton shapes that outline the incoming content rather than a centred spinner.
- [ ] Ask users whether a scrollable list has more items; a no is a closure failure.

ANTI-PATTERN VS REMEDIATION

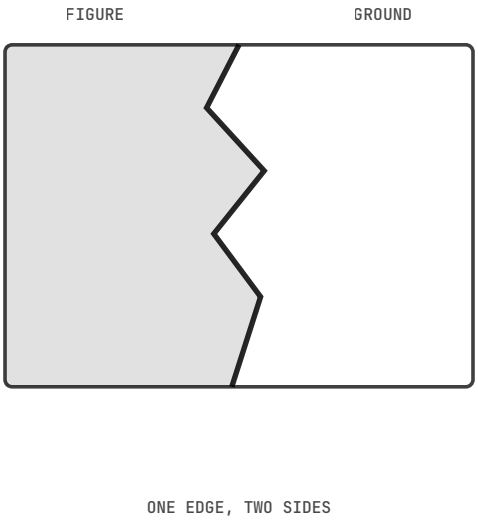
Violation: A carousel snaps flush to the margins, so the row reads as complete and nobody swipes.

Fix: One card sits two-thirds off the edge, so the row visibly continues.

Source: Kurt Koffka (1935). *Principles of Gestalt Psychology.* Harcourt, Brace & World.

Law of Figure/Ground

A view reads as a figure against a ground, and one edge separates them.



WHEN TO APPLY

A dialog opens with no dimmed backdrop, so its text bleeds into the page behind it.

GES-06

Law of Figure/Ground

OPERATIONAL CHECKLIST

- [ ] Dim the page under a modal so the dialog is the only thing competing for the eye.
- [ ] Raise menus and sheets off the surface with one shadow direction, not several.
- [ ] Keep body text at least 7:1 against its background; the figure needs its ground.
- [ ] Check a screenshot in grey; if figure and ground merge, contrast is carrying it alone.

ANTI-PATTERN VS REMEDIATION

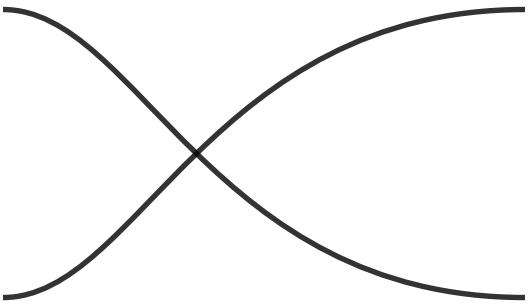
Violation: A modal opens over a dense page with no scrim, and its copy mixes with the text behind it.

Fix: The modal sits on a dark scrim with one soft shadow, and the page behind recedes.

Source: Edgar Rubin (1915). *Synsoplevede Figurer.* Gyldendalske Boghandel.

Law of Continuity (Good Continuation)

Marks along one smooth path read as a single line, not a series.



EACH LINE CONTINUES

WHEN TO APPLY

Numbers in a table centre-align, so the decimal points stagger down the column.

GES-07

Law of Continuity (Good Continuation)

OPERATIONAL CHECKLIST

- [ ] Right-align numeric columns and use tabular figures so digits stack in one plane.
- [ ] Run onboarding steps along one visible axis instead of wrapping them into a grid.
- [ ] Keep scan paths predictable; do not break a column of links with a full-width banner.
- [ ] Track a reader's eye across a dense screen; sideways jumps mark a broken continuation.

ANTI-PATTERN VS REMEDIATION

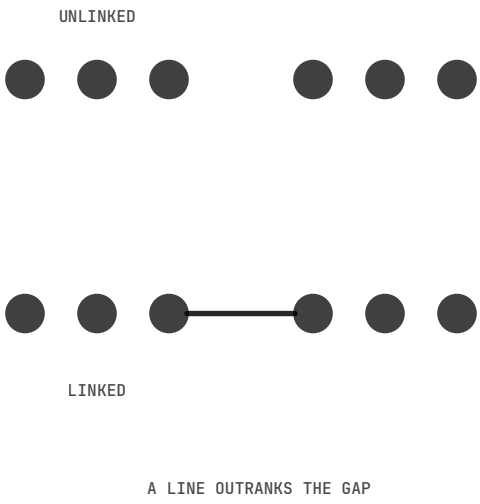
Violation: A ledger centres its figures, the decimal points stagger, and the column cannot be scanned.

Fix: Every amount is right-aligned in tabular figures, forming one vertical line of decimals.

Source: Max Wertheimer (1923). *Untersuchungen zur Lehre von der Gestalt II.* Psychologische Forschung, 4, 301-350.

Law of Uniform Connectedness

Where a line joins two things, people group them before they weigh the distance.



A LINE OUTRANKS THE GAP

WHEN TO APPLY

Wizard steps float unconnected, so the order has to be inferred from the numbers.

GES-08

Law of Uniform Connectedness

OPERATIONAL CHECKLIST

- [ ] Join nodes in a flow editor with one continuous stroke, not arrows that stop short.
- [ ] Connect checkout steps with a single track so the sequence is drawn, not implied.
- [ ] Attach a tab to its panel with a shared border instead of relying on colour alone.
- [ ] Remove the connecting line as a test; if grouping survives, the line was decoration.

ANTI-PATTERN VS REMEDIATION

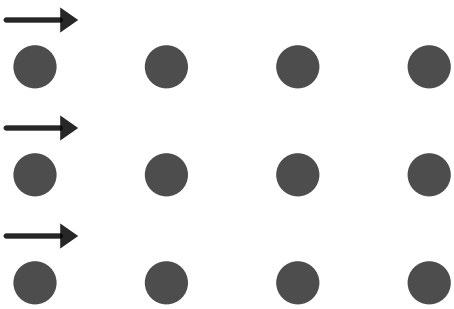
Violation: Four numbered circles float with no links, and the user must guess the sequence.

Fix: The circles sit on one continuous track that fills as each stage completes.

Source: Stephen E. Palmer & Irvin Rock (1994). *Rethinking perceptual organization: The role of uniform connectedness.* Psychonomic Bulletin & Review, 1(1), 29-55.

Law of Common Fate

Things that move together in one direction are seen as one thing, whatever else differs.



SAME DIRECTION, ONE THING

WHEN TO APPLY

A sidebar collapses while its labels reflow at a different speed, so the panel tears.

GES-09

Law of Common Fate

OPERATIONAL CHECKLIST

- [ ] Animate a menu and its items as one surface, not as a stagger of separate parts.
- [ ] Give every element of a collapsing panel the same duration and easing curve.
- [ ] Dismiss table rows along one exit path so the remaining rows move as a block.
- [ ] Record a slowed transition; any part that arrives late is moving in its own direction.

ANTI-PATTERN VS REMEDIATION

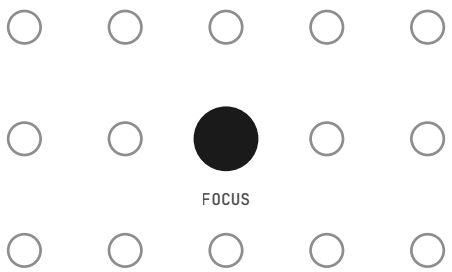
Violation: A sidebar slides left while its child text wraps and drops, so the panel comes apart.

Fix: Sidebar, labels, and icons translate together on one 200 ms curve.

Source: Max Wertheimer (1923). *Untersuchungen zur Lehre von der Gestalt II.* Psychologische Forschung, 4, 301-350.

Law of Focal Point

One mark with the most contrast takes the eye, and everything else waits its turn.



ONE MARK TAKES THE EYE

WHEN TO APPLY

Every plan in a pricing table has a solid red Buy button, so none stands out.

GES-10

Law of Focal Point

OPERATIONAL CHECKLIST

- [ ] Give one action per view the strongest fill, and demote the rest to outline.
- [ ] Allow one high-contrast badge per screen; a second cancels the first.
- [ ] Put the recommended option in one accent border rather than five coloured buttons.
- [ ] Screenshot the view, blur it, and see which element survives; it should be the action.

ANTI-PATTERN VS REMEDIATION

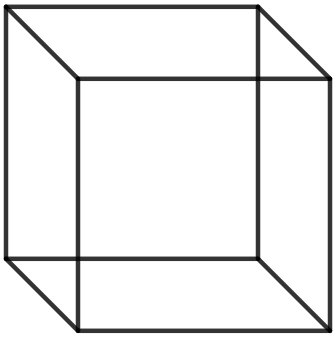
Violation: Four pricing cards each carry an identical solid red Buy button, so no tier is focal.

Fix: Three tiers use outline buttons; the recommended one is framed and filled.

Source: Stephen E. Palmer (1999). *Vision Science: Photons to Phenomenology.* MIT Press.

Law of Past Experience

People read an ambiguous mark by the conventions they already carry.



SAME EDGES, TWO READINGS

WHEN TO APPLY

A green button deletes an account, or a back arrow sits on the right of the row.

GES-11

Law of Past Experience

OPERATIONAL CHECKLIST

- [ ] Keep red for destructive, green for success, and amber for caution, because users already do.
- [ ] Use a cart for the basket, a gear for settings, a speech bubble for comments.
- [ ] Put forward on the right and back on the left in left-to-right locales.
- [ ] Test an icon without its label; a wrong guess marks a convention the user does not share.

ANTI-PATTERN VS REMEDIATION

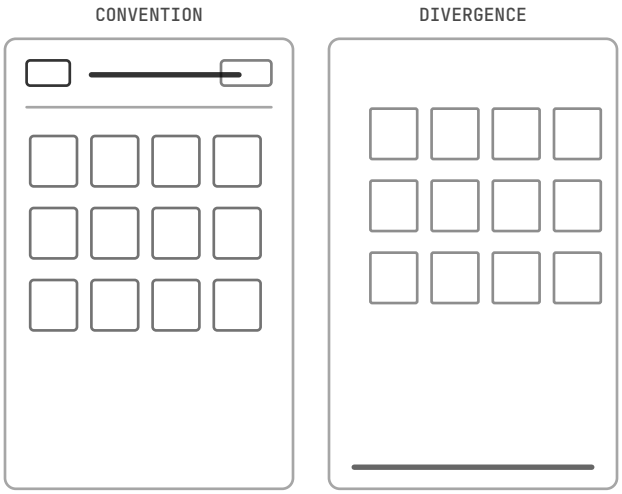
Violation: A green button deletes an account, and the undo prompt is the first time anyone is told.

Fix: Destructive actions are red and named; success is green and never destroys.

Source: Kurt Koffka (1935). *Principles of Gestalt Psychology.* Harcourt, Brace & World.

Jakob's Law

Users spend most time on other interfaces, expecting yours to function identically.



FAMILIAR GRAMMAR BECOMES EXPECTATION

WHEN TO APPLY

Users hesitate, hunt for navigation, or misuse controls because the layout diverges from conventions they know from other products.

HMM-01

Jakob's Law

OPERATIONAL CHECKLIST

- [ ] Anchor the logo top-left, global navigation top or left, and account controls top-right.
- [ ] Keep checkout steps in the expected billing, shipping, and payment order.
- [ ] Preserve standard keyboard shortcuts and browser history behavior; never hijack them.

ANTI-PATTERN VS REMEDIATION

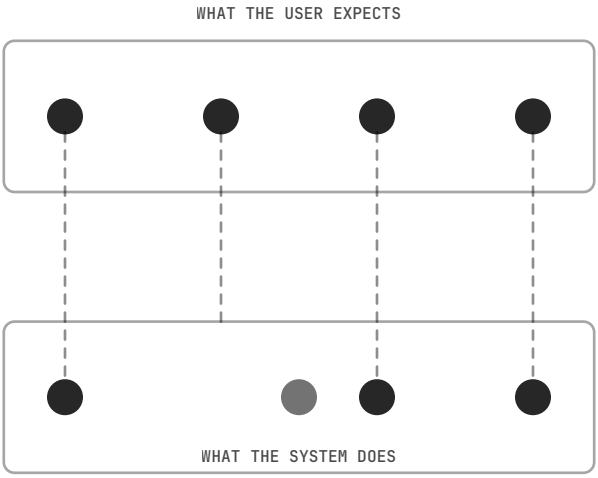
Violation: A web app rebinds the browser back button to internal tab switching and buries global search in the footer.

Fix: Call history.pushState or replaceState for back entries; keep search in a persistent header input focused with /, a key the page receives before the browser's quick-find default, not Cmd/Ctrl+K, which the browser reserves as a command.

Source: Jakob Nielsen (2000). *Designing Web Usability: The Practice of Simplicity.* New Riders Publishing.

Mental Model Theory

People act on a simplified internal model of a system, not the system itself.



WHERE THE MODEL AND THE SYSTEM PART

WHEN TO APPLY

Users invent their own account of how a feature works, then blame the interface when that account predicts the wrong result.

HMM-02

Mental Model Theory

OPERATIONAL CHECKLIST

- [ ] Name folders, tags, and documents after the objects users already handle in their work.
- [ ] Update the status view from the same event that performed the action, not from a poll.
- [ ] Keep storage terms such as rows, records, and foreign keys out of user-facing labels.
- [ ] Ask users to predict what an action will do before they use it, and record each wrong prediction.

ANTI-PATTERN VS REMEDIATION

Violation: An email client labels a delete action with a database operation, so saving a draft looks like destroying a record.

Fix: Delete moves the message to a Trash folder and states that it stays recoverable for 30 days before purging.

Source: Kenneth Craik (1943). *The Nature of Explanation.* Cambridge University Press.

Conceptual Integrity

Every module should express the same set of design ideas in the same way.

HEADER, BODY, ACTION EVERY TIME



ONE GRAMMAR, THREE CONTAINERS

WHEN TO APPLY

Two screens in one product use different labels, layouts, or gestures for the same job, so each has to be learned from scratch.

HMM-03

Conceptual Integrity

OPERATIONAL CHECKLIST

- [ ] Open the same kind of record the same way everywhere: one dialog, one drawer, or one page.
- [ ] Give one permission level one verb across the product; grep the interface strings for synonyms.
- [ ] Render page headers from one component so type, spacing, and breadcrumbs cannot drift apart.

ANTI-PATTERN VS REMEDIATION

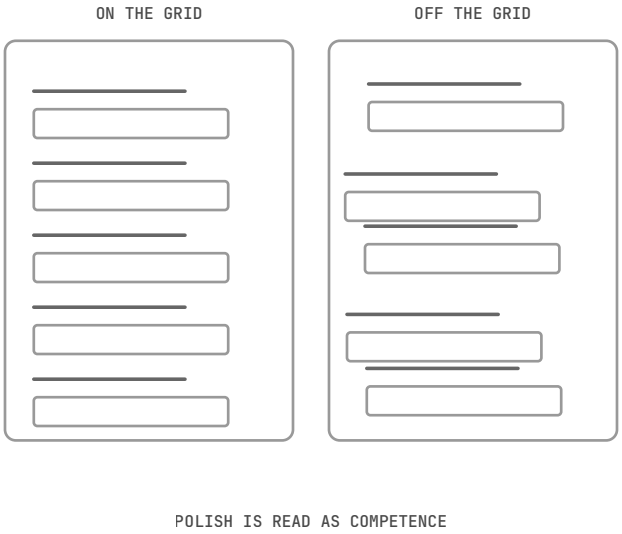
Violation: A dashboard creates a project in a modal, a task on a new page, and a client in a drawer.

Fix: All record creation opens the same slide-over sheet, with the same button order and unsaved-changes guard.

Source: Frederick P. Brooks Jr. (1975). *The Mythical Man-Month: Essays on Software Engineering.* Addison-Wesley.

Aesthetic-Usability Effect

Polished interfaces are judged more usable than plain ones that do the same work.



WHEN TO APPLY

A restyle changes only color, type, and spacing; testers still rate the product easier to use and stop noticing its flaws.

HMM-04

Aesthetic-Usability Effect

OPERATIONAL CHECKLIST

- [ ] Derive every margin, padding, and gap from one 4 or 8 pixel scale; no arbitrary values.
- [ ] Align text baselines and control edges to shared guides, and check the page at 100 percent zoom.
- [ ] Draw border radii, shadow levels, and neutral colors from one token file.
- [ ] Test perceived usability separately from task completion; a high aesthetic score is not task success.

ANTI-PATTERN VS REMEDIATION

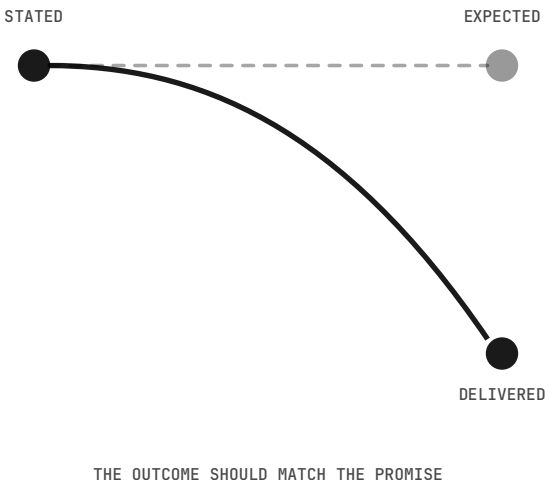
Violation: An analytics product ships default browser buttons, misaligned table columns, and unstyled alert text.

Fix: It uses tabular figures, one radius scale, and a neutral ramp, with spacing tokens shared across the product.

Source: Masaaki Kurosu & Kaori Kashimura (1995). *Apparent usability vs. inherent usability.* CHI '95 Conference Companion, 292-293.

Principle of Least Astonishment (POLA)

A control should do what its appearance and label lead people to expect.



WHEN TO APPLY

A familiar control does something unexpected: Save clears the form, a filter reloads the page.

HMM-05

Principle of Least Astonishment (POLA)

OPERATIONAL CHECKLIST

- [ ] Keep a control's label and function stable across hover, focus, and click; never swap its action.
- [ ] Preserve typed values when server validation fails and mark the failing fields beside them.
- [ ] Give destructive actions an undo window or a confirmation that names the object being removed.
- [ ] Record where users hesitate or backtrack after an action, and treat each as a violated expectation.

ANTI-PATTERN VS REMEDIATION

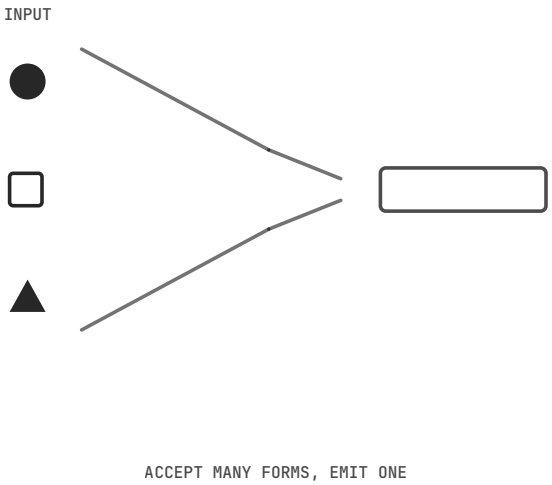
Violation: Pressing Save Draft clears every field and shows no confirmation, so the draft exists in local storage or nowhere.

Fix: Save Draft shows a saved-at timestamp, keeps the fields filled, and syncs in the background.

Source: M. F. Cowlishaw (1984). *The design of the REXX language.* IBM Systems Journal, 23(4), 326-335.

Postel's Law (Robustness Principle)

Accept input in the forms people send; emit output in one strict form.



ACCEPT MANY FORMS, EMIT ONE

WHEN TO APPLY

Valid input is rejected over formatting, or downstream services each get a different shape.

HMM-06

Postel's Law (Robustness Principle)

OPERATIONAL CHECKLIST

- [ ] Strip spaces, hyphens, and parentheses from phone and card fields before validating them.
- [ ] Parse dates against an explicit list of accepted formats and show those forms beside the field.
- [ ] Emit one canonical format from the API boundary and validate the payload against a schema.

ANTI-PATTERN VS REMEDIATION

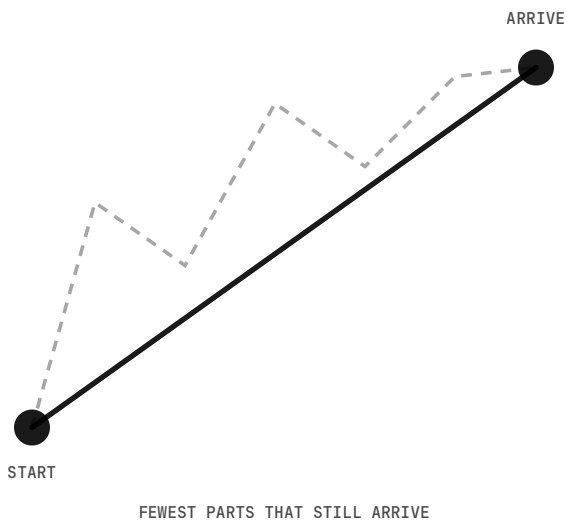
Violation: Checkout rejects a card number because the user typed spaces between the digit groups, and the error says invalid characters.

Fix: The field accepts any delimiter, removes non-digits, and displays the number in grouped blocks as it is typed.

Source: Jon Postel (1980). *DoD standard Transmission Control Protocol.* IETF RFC 761.

Occam's Razor (Principle of Parsimony)

Prefer the design that explains everything with the fewest added assumptions.



WHEN TO APPLY

Options and edge-case branches pile up until nobody can describe the main path.

HMM-07

Occam's Razor (Principle of Parsimony)

OPERATIONAL CHECKLIST

- [ ] Delete gradients, dividers, and wrapper elements that carry no content and no state.
- [ ] Render one primary action on a transactional screen and demote the rest below it.
- [ ] Merge Filter, Sort, and Search into one query bar when they act on the same collection.
- [ ] Time first-time users on the main task before and after a simplification, not just their opinion of it.

ANTI-PATTERN VS REMEDIATION

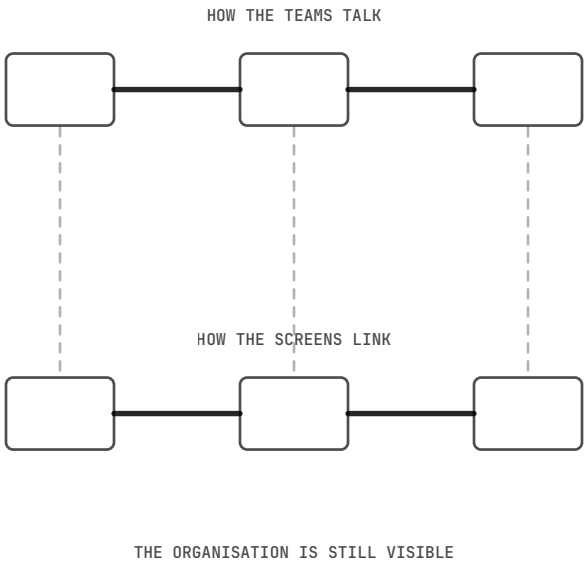
Violation: An editor shows 45 toolbar buttons for rare typographic cases, so new writers cannot find bold or undo.

Fix: A bare canvas appears first, with a contextual toolbar only while text is selected.

Source: John Maeda (2006). *The Laws of Simplicity.* MIT Press.

Conway's Law

A system's structure ends up copying the communication structure of the teams that built it.



WHEN TO APPLY

Two teams own half a flow each, and the product asks users to cross a seam no user would draw.

HMM-08

Conway's Law

OPERATIONAL CHECKLIST

- [ ] Organise menus by the user's jobs, not by the departments that own the backend services.
- [ ] Share one set of design tokens across services so separately built screens look related.
- [ ] Merge settings the back end splits across services behind one account screen.
- [ ] Before a reorganisation ships, map its team boundaries onto the screens users must cross.

ANTI-PATTERN VS REMEDIATION

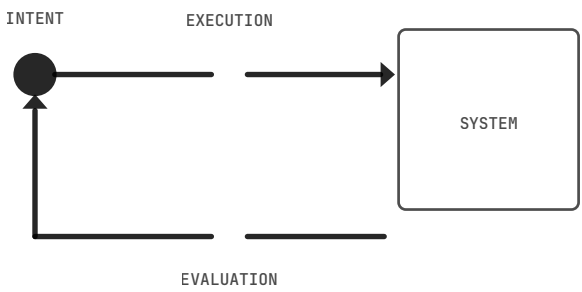
Violation: A telecom portal sends customers to three subdomains with different themes to manage phone, broadband, and billing.

Fix: One identity and one design shell put all three offerings in a single services dashboard.

Source: Melvin E. Conway (1968). *How Do Committees Invent?* Datamation, 14(4), 28-31.

Gulf of Execution and Gulf of Evaluation

Keep intent, action, and visible result close enough that people never have to guess.



TWO GAPS: TO ACT, AND TO SEE

WHEN TO APPLY

A button gives no sign it worked, so people click again and create duplicate orders.

HMM-09

Gulf of Execution and Gulf of Evaluation

OPERATIONAL CHECKLIST

- [ ] Mark clickable elements with a signifier users can see before they hover or click.
- [ ] Change the control and show a status line as soon as an action is accepted.
- [ ] State in the error message what happened, which object it concerned, and the next step.
- [ ] Ask users to describe what the system did after each action; treat a wrong answer as a gulf.

ANTI-PATTERN VS REMEDIATION

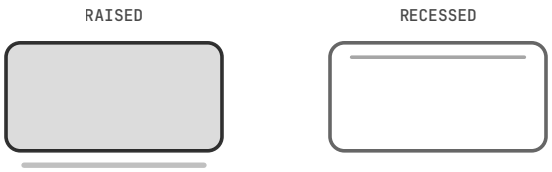
Violation: Clicking Process leaves the cursor unchanged and shows nothing, so the operator clicks three times and files three orders.

Fix: The button disables itself, shows a spinner, and relabels to Processing until the result appears.

Source: Donald A. Norman (1986). *Cognitive Engineering.* User Centered System Design, 31-62.

Affordance Theory

An object's properties suggest what can be done with it, before any label says so.



EITHER SURFACE CAN SAY PRESS ME

WHEN TO APPLY

A control that works looks exactly like the text around it, so nobody tries it.

HMM-10

Affordance Theory

OPERATIONAL CHECKLIST

- [ ] Give every control a fill, border, or weight the surrounding text does not have.
- [ ] Keep links underlined or coloured at rest; hover styling alone is not a signifier.
- [ ] Draw a grip on anything draggable, and make the whole surface respond to the drag.
- [ ] Watch first-time users try to click; every dead click marks an affordance that was not visible.

ANTI-PATTERN VS REMEDIATION

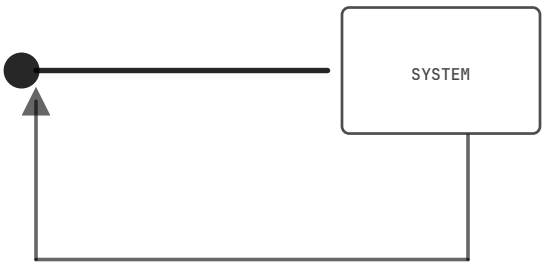
Violation: Call-to-action buttons are set as plain text with no border, fill, or underline, identical to body copy.

Fix: Controls sit in filled rounded containers that contrast with the surface and lift on hover.

Source: James J. Gibson (1977). *The Theory of Affordances.* Perceiving, Acting, and Knowing (Shaw & Bransford, eds.).

Feedback Principle (Cybernetic Loop)

People can steer only what reports back what their last action did.



AN ACTION THAT COMES BACK

WHEN TO APPLY

A long upload shows nothing until it finishes, so people reload and start it again.

HMM-11

Feedback Principle (Cybernetic Loop)

OPERATIONAL CHECKLIST

- [ ] Show a state change on every control as soon as it is pressed, not when the request returns.
- [ ] Keep a progress indicator for any operation that outlasts a second or two.
- [ ] Report the outcome of a save where the user is looking, not only in a toast that fades.
- [ ] Measure how long the interface takes to acknowledge an action, and treat silence as a defect.

ANTI-PATTERN VS REMEDIATION

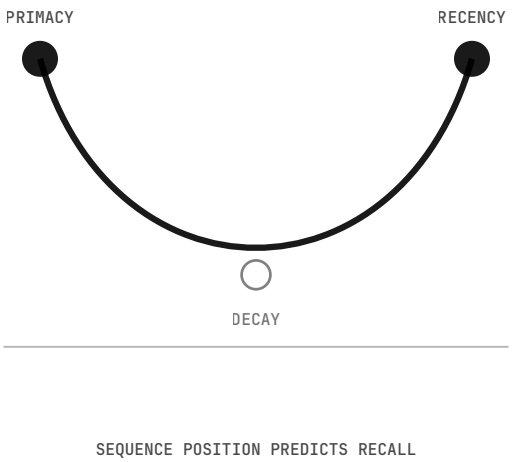
Violation: A file drop zone looks the same while a file is dragged over it, so the user cannot tell whether a drop will be accepted.

Fix: The zone outlines itself, tints its surface, and shows Drop files to upload while a file is over it.

Source: Norbert Wiener (1948). *Cybernetics: Or Control and Communication in the Animal and the Machine.* MIT Press.

Serial Position Effect

Humans recall items positioned at the beginning and end of sequences most reliably.



WHEN TO APPLY

Critical navigation or actions sit mid-list and are overlooked while first and last items dominate attention.

MPA-01

Serial Position Effect

OPERATIONAL CHECKLIST

- [ ] Put critical destinations at the far left or top and far right or bottom of navigation.
- [ ] Place core value propositions in the first rows and conversion actions in the last.
- [ ] Pin frequent dropdown options to the top and summary categories to the bottom.

ANTI-PATTERN VS REMEDIATION

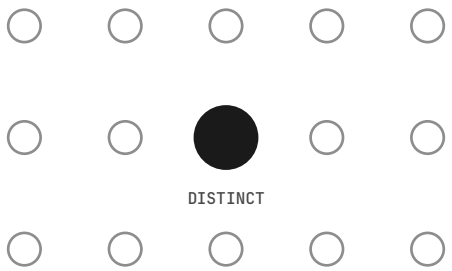
Violation: A 12-item flat navigation menu buries the primary analytics dashboard in the middle.

Fix: Group the 12 items into five to seven top-level destinations (DEC-02), then put Dashboard first and Settings last.

Source: Bennet B. Murdock Jr. (1962). *The Serial Position Effect of Free Recall.* Journal of Experimental Psychology, 64(5), 482-488.

Von Restorff Effect (Isolation Effect)

An item that differs from its neighbors is remembered better than the items around it.



ONE ITEM STANDS OUT AND IS REMEMBERED

WHEN TO APPLY

One row, button, or price is remembered while identical neighbors blur together.

MPA-02

Von Restorff Effect (Isolation Effect)

OPERATIONAL CHECKLIST

- [ ] Emphasize one item per view; every extra highlight competes with the one meant to stand out.
- [ ] Match the emphasis to the decision: the row people must recall gets the weight.
- [ ] Test recall a day later, when the standout item is what remains.

ANTI-PATTERN VS REMEDIATION

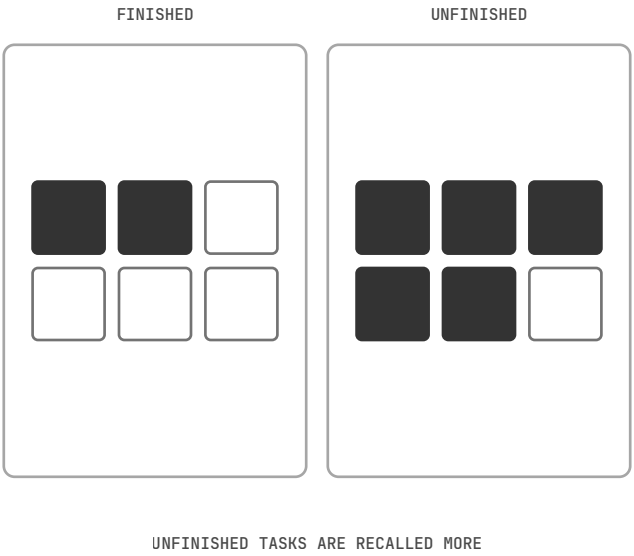
Violation: Every plan card is bright blue, so none of the three is remembered.

Fix: Only the recommended plan is filled; the other two stay outlined.

Source: Hedwig von Restorff (1933). *Über die Wirkung von Bereichsbildungen im Spurenfeld.* Psychologische Forschung, 18(1), 299-342.

Zeigarnik Effect

Unfinished tasks stay in memory longer than tasks already completed.



WHEN TO APPLY

People return days later and remember exactly where they stopped.

MPA-03

Zeigarnik Effect

OPERATIONAL CHECKLIST

- [ ] Save progress automatically and offer to resume where the person stopped.
- [ ] Show unfinished items on return; do not make people hunt for them.
- [ ] End a session with a visible next step, not a blank confirmation.

ANTI-PATTERN VS REMEDIATION

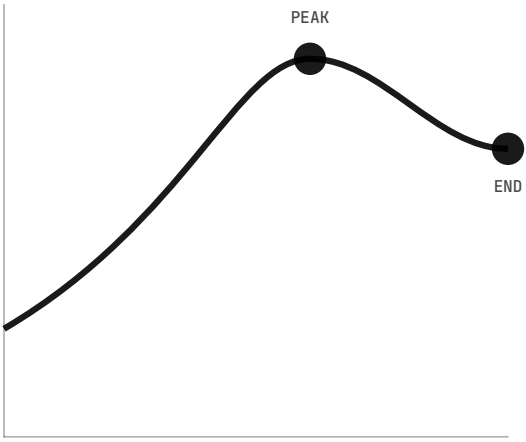
Violation: A half-finished application vanishes on reload, so the person starts over.

Fix: The draft is saved and the return screen names the next question.

Source: Bluma Zeigarnik (1927). *Über das Behalten von erledigten und unerledigten Handlungen.* Psychologische Forschung, 9.

Peak-End Rule

An experience is remembered by its peak and its ending, not its average.



MEMORY FOLLOWS THE PEAK AND THE END

WHEN TO APPLY

A smooth session is rated harshly because its last screen failed.

MPA-04

Peak-End Rule

OPERATIONAL CHECKLIST

- [ ] End every flow on a clear confirmation that names what happens next.
- [ ] Soften the most intense step: the peak sets the rating more than the middle.
- [ ] Fix mid-flow defects on their own terms: a strong ending recolors the memory, it does not repair the experience.

ANTI-PATTERN VS REMEDIATION

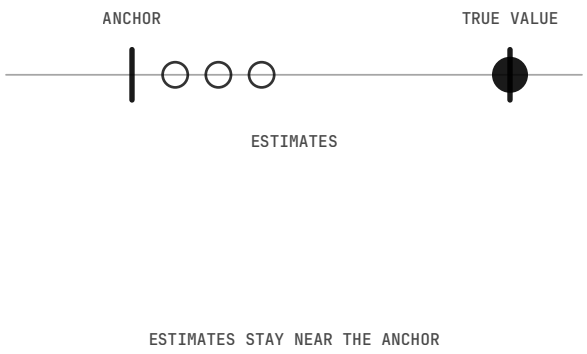
Violation: A fast checkout ends on a paywall error, so the whole flow is remembered as bad.

Fix: The last screen thanks the person and confirms the order.

Source: Daniel Kahneman, Barbara L. Fredrickson, Charles A. Schreiber & Donald A. Redelmeier (1993). *When More Pain Is Preferred to Less: Adding a Better End.* Psychological Science, 4(6), 401-405.

Anchoring Bias

An initial number pulls later estimates toward it, even when it is arbitrary.



WHEN TO APPLY

A suggested price, quantity, or budget shifts what people choose.

MPA-05

Anchoring Bias

OPERATIONAL CHECKLIST

- [ ] Remove arbitrary defaults before asking for an estimate.
- [ ] Show a reference range the evidence supports, not a number near the answer.
- [ ] Test the same question with and without a starting figure.

ANTI-PATTERN VS REMEDIATION

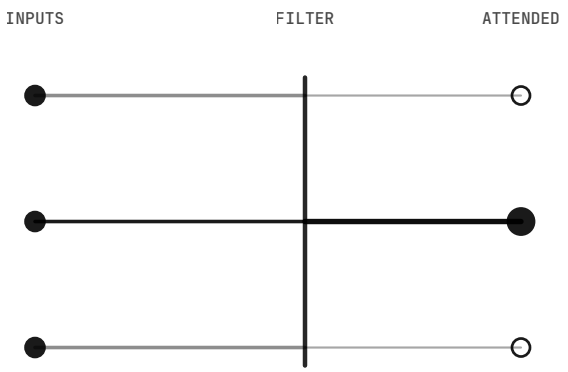
Violation: A donation field prefilled with the highest tier lifts every gift.

Fix: The field starts empty and offers a range with its basis.

Source: Amos Tversky & Daniel Kahneman (1974). *Judgment under Uncertainty: Heuristics and Biases.* Science, 185(4157), 1124-1131.

Selective Attention and Attentional Bottleneck

Attention admits one channel at a time; unattended channels survive only as fragments.



ONE CHANNEL PASSES; THE OTHERS ARRIVE WEAKENED

WHEN TO APPLY

Two simultaneous requests compete, and the second is missed entirely.

MPA-06

Selective Attention and Attentional Bottleneck

OPERATIONAL CHECKLIST

- [ ] Route interruptions to one channel instead of stacking toasts and modals.
- [ ] Do not put two demands in the same moment; queue the second.
- [ ] Keep a missed request recoverable where the person will look for it.

ANTI-PATTERN VS REMEDIATION

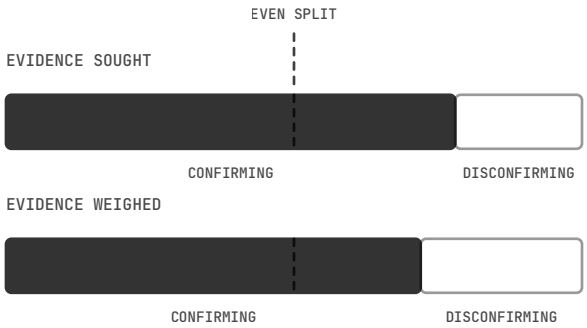
Violation: A toast, a modal, and a banner ask for attention at once.

Fix: One request at a time, queued in a place the person can revisit.

Source: Donald E. Broadbent (1958). *Perception and Communication.* Pergamon Press, London.

Confirmation Bias

People seek and weight evidence that confirms what they already believe.



CONFIRMING EVIDENCE GETS THE FIRST LOOK

WHEN TO APPLY

A dashboard shows only the runs that support the current plan.

MPA-07

Confirmation Bias

OPERATIONAL CHECKLIST

- [ ] Show disconfirming results at the same prominence as confirming ones.
- [ ] Ask what evidence would change the decision before the test runs.
- [ ] Record every run, including the ones that refute the hypothesis.

ANTI-PATTERN VS REMEDIATION

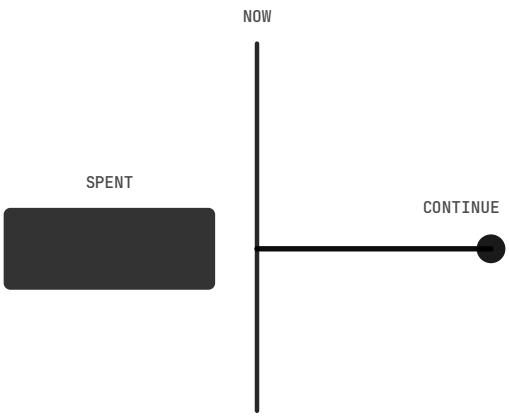
Violation: Only successful test runs are shown, so a failing design looks proven.

Fix: Report all runs, with refutations beside the successes.

Source: Raymond S. Nickerson (1998). *Confirmation Bias: A Ubiquitous Phenomenon in Many Guises.* Review of General Psychology, 2(2), 175-220.

Sunk Cost Fallacy

Money already spent pulls the next decision, though only future costs should.



WHAT IS SPENT SHOULD NOT STEER WHAT COMES NEXT

WHEN TO APPLY

A team keeps funding a feature because of what it has already cost.

MPA-08

Sunk Cost Fallacy

OPERATIONAL CHECKLIST

- [ ] Name the cost already spent and leave it out of the go/no-go calculation.
- [ ] Judge the next step by what it costs from here and what it returns.
- [ ] Rewrite a stuck decision as if nothing had been spent yet, then compare.

ANTI-PATTERN VS REMEDIATION

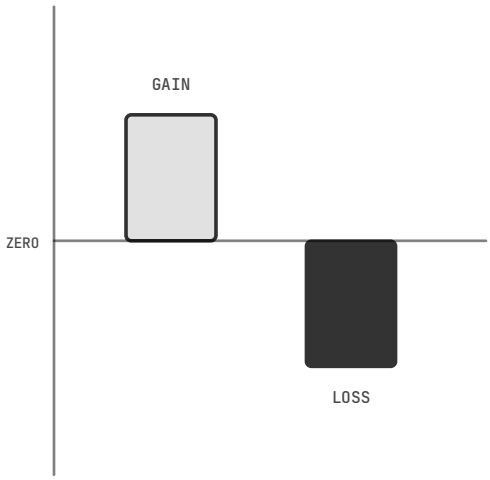
Violation: A failing feature keeps its budget because half of it is already built.

Fix: Decide on the remaining cost and return; note the spent money and exclude it.

Source: Hal R. Arkes & Catherine Blumer (1985). *The psychology of sunk cost.* Organizational Behavior and Human Decision Processes, 35(1), 124-140.

Loss Aversion

Losses weigh on choices more than equivalent gains do.



A LOSS OUTWEIGHS THE SAME-SIZED GAIN

WHEN TO APPLY

A change is refused for what it might lose, not weighed by what it gains.

MPA-09

Loss Aversion

OPERATIONAL CHECKLIST

- [ ] Show the loss and the gain of a change side by side, in the same units.
- [ ] Describe what staying costs as plainly as what moving costs.
- [ ] Test whether the framing moved the decision, not only the preference.

ANTI-PATTERN VS REMEDIATION

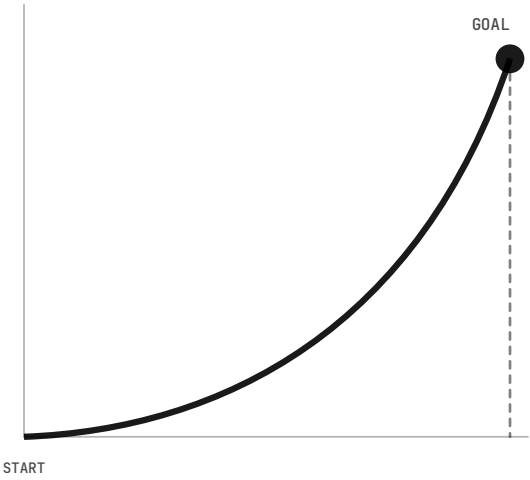
Violation: A migration is presented only as what could break, so nobody approves it.

Fix: State the risk and the gain together, with the same visibility and the same units.

Source: Daniel Kahneman & Amos Tversky (1979). *Prospect Theory: An Analysis of Decision under Risk.* Econometrica, 47(2), 263-291.

Goal-Gradient Effect

Effort rises as the goal nears; the last stretch gets the most.



EFFORT RISES AS THE GOAL NEARS

WHEN TO APPLY

People start slowly on a long task and speed up near its end.

MPA-10

Goal-Gradient Effect

OPERATIONAL CHECKLIST

- [ ] Show how far the person has come, not only how much remains.
- [ ] Split a long run into visible goals, so a near one is always in reach.
- [ ] Put the last review beside the finish, where the end is in view.

ANTI-PATTERN VS REMEDIATION

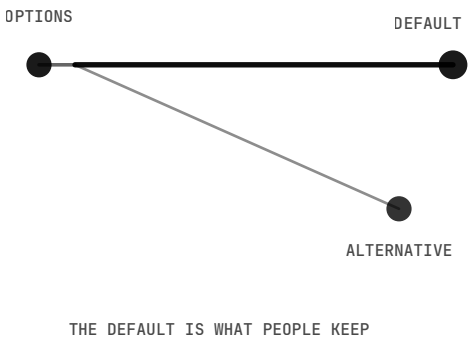
Violation: A ten-step setup shows one spinner for the whole run, so people leave at step three.

Fix: Show each completed step and the count remaining, so the goal stays in view.

Source: Ran Kivetz, Oleg Urminsky & Yuhuang Zheng (2006). *The Goal-Gradient Hypothesis Resurrected: Purchase Acceleration, Illusionary Goal Progress, and Customer Retention.* Journal of Marketing Research, 43(1), 39-58.

Status Quo Bias

People keep the option they already have unless something forces a change.



THE DEFAULT IS WHAT PEOPLE KEEP

WHEN TO APPLY

A setting nobody changed is treated as the one people chose.

MPA-11

Status Quo Bias

OPERATIONAL CHECKLIST

- [ ] Name the default in words and say what happens if nobody chooses.
- [ ] Make any move away from the default reversible in one step.
- [ ] Give the alternatives the same visibility as the default; do not bury them.

ANTI-PATTERN VS REMEDIATION

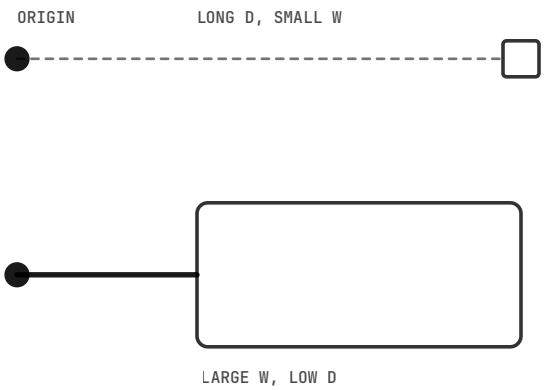
Violation: Opt-out sits on a second screen, so the default is the only path anyone finds.

Fix: Show the default and the alternative on one screen, equally visible, both one click away.

Source: William Samuelson & Richard Zeckhauser (1988). *Status quo bias in decision making.* Journal of Risk and Uncertainty, 1(1), 7-59.

Fitts's Law

Target acquisition time is a function of distance and target bounding width.



$$T = a + b \log_2(1 + D/W)$$

WHEN TO APPLY

Users miss small controls or take long pointer trips for frequent actions, especially on touch screens.

TME-01

Fitts's Law

OPERATIONAL CHECKLIST

- [ ] Make touch targets at least 44 x 44 px, or 24 x 24 px with adequate spacing.
- [ ] Pin frequent primary controls to screen edges or corners on desktop.
- [ ] Place contextual actions next to the initiating click point.

ANTI-PATTERN VS REMEDIATION

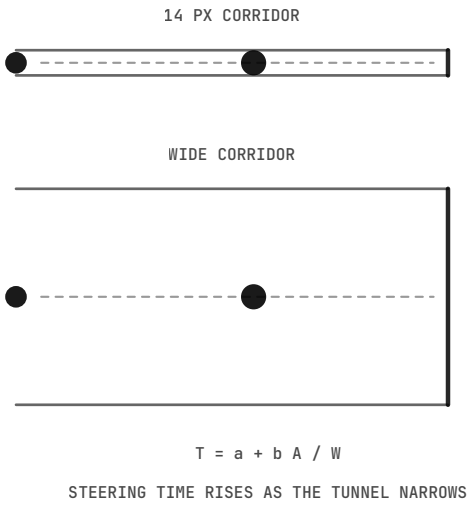
Violation: Mobile checkout puts a 12 x 12 px Confirm Order text link in the extreme top-right corner.

Fix: Use a full-width sticky footer button at least 48 px tall within thumb reach.

Source: Paul M. Fitts (1954). *The Information Capacity of the Human Motor System in Controlling the Amplitude of Movement.* Journal of Experimental Psychology, 47(6), 381-391.

Accot-Zhai Steering Law

Steering time through a tunnel grows with its length and shrinks with its width.



WHEN TO APPLY

A cursor entering a nested fly-out drifts a few pixels sideways and the menu collapses.

TME-02

Accot-Zhai Steering Law

OPERATIONAL CHECKLIST

- [ ] Replace narrow nested fly-outs with a flat mega-menu or a wide hover corridor.
- [ ] Add a triangular safe-hover zone between the pointer and the open submenu.
- [ ] Keep generous padding on drag tracks, sliders, and scrollbar thumbs.

ANTI-PATTERN VS REMEDIATION

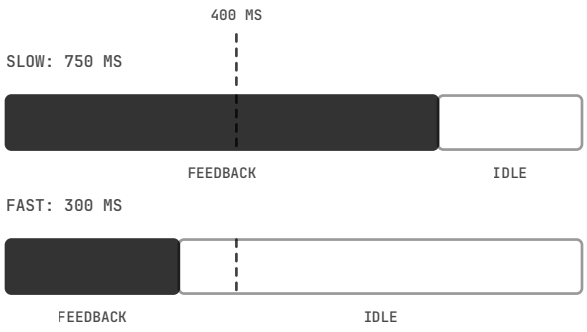
Violation: A desktop menu makes the pointer cross a fourteen-pixel submenu path, and one pixel outside closes it.

Fix: An invisible hit polygon joins the pointer to the open submenu, so a diagonal move keeps it open.

Source: Johnny Accot & Shumin Zhai (1997). *Beyond Fitts' Law: Models for Trajectory-Based HCI Tasks.* ACM CHI '97 Proceedings, 295-302.

Doherty Threshold

When system response drops under four hundred milliseconds, people stay engaged and work faster.



RESPONSE UNDER 400 MS HOLDS ATTENTION

WHEN TO APPLY

A click blocks the interface for most of a second, and the person clicks again or looks away.

TME-03

Doherty Threshold

OPERATIONAL CHECKLIST

- [ ] Update the interface at once on input, then reconcile with the server in the background.
- [ ] Show skeletons that match the final layout when a wait exceeds four hundred milliseconds.
- [ ] Prefetch likely next screens on hover or focus.

ANTI-PATTERN VS REMEDIATION

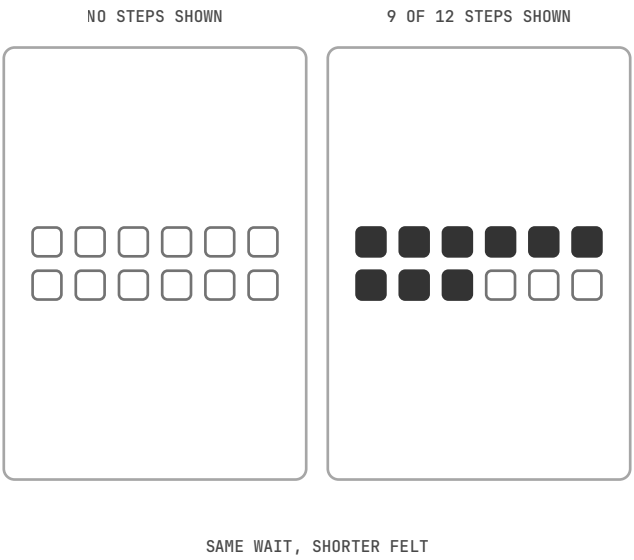
Violation: A like button freezes for 750 ms while the request runs, with no visible change.

Fix: The icon fills immediately, the counter increments locally, and the request reconciles later.

Source: Walter J. Doherty & Ahrvind J. Thadhani (1982). *The Economic Value of Rapid Response Time.* IBM Technical Report GE20-0752-0, White Plains, NY.

Perceived Performance Heuristics

Progress cues make the same wait feel shorter than a blank screen does.



WHEN TO APPLY

An export runs for twelve seconds behind a blank screen, and people refresh because they think it froze.

TME-04

Perceived Performance Heuristics

OPERATIONAL CHECKLIST

- [ ] Pair long operations with a determinate bar and motion that accelerates toward the end.
- [ ] Replace a static spinner with skeleton screens or a pulsing placeholder.
- [ ] Name the current stage in plain words: connecting, encrypting, finishing.

ANTI-PATTERN VS REMEDIATION

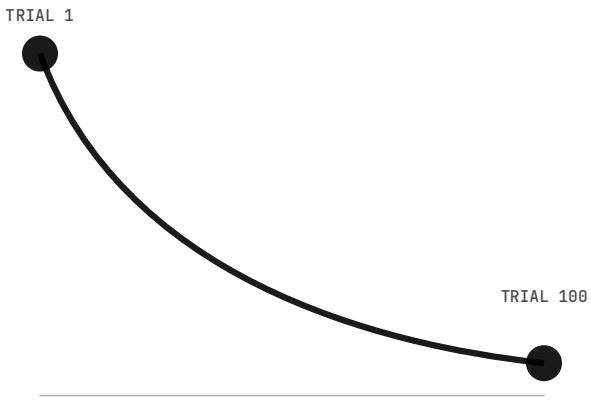
Violation: A large export shows a blank page and a static cursor for twelve seconds.

Fix: A progress dialog shows a determinate bar, the current stage, and an estimated remainder.

Source: Steven C. Seow (2008). *Designing and Engineering Time: The Psychology of Time Perception in Software.* Addison-Wesley.

Power Law of Practice

Task time falls as a power of practice repetitions: steep early gains, then diminishing returns.



STEEP EARLY GAINS, THEN DIMINISHING RETURNS

WHEN TO APPLY

A workflow someone uses every day takes as long in month six as it did in week one.

TME-05

Power Law of Practice

OPERATIONAL CHECKLIST

- [ ] Keep frequent actions in one fixed place so repetition trains aim instead of search.
- [ ] Give daily paths keyboard shortcuts and let the interface remember recent choices.
- [ ] Make the fastest route the default, not something found in settings.
- [ ] Measure time-on-task for repeat workflows; a flat curve means the interface blocks practice.

ANTI-PATTERN VS REMEDIATION

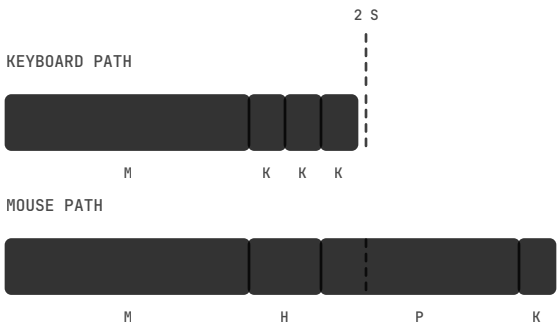
Violation: A ticketing tool moves its main action each release, so agents relearn the target every quarter.

Fix: The primary action keeps one position and gains a shortcut once an agent repeats it.

Source: Allen Newell & Paul S. Rosenbloom (1981). *Mechanisms of Skill Acquisition and the Law of Practice.* In J. R. Anderson (Ed.), Cognitive Skills and Their Acquisition, Lawrence Erlbaum Associates, Hillsdale, NJ.

Keystroke-Level Model

Expert task time sums discrete operator durations: keystroke, pointing, homing, mental, and response.



TOTAL TIME IS THE SUM OF OPERATORS

WHEN TO APPLY

A task feels slow although no single step is slow; the time goes into small transitions.

TME-06

Keystroke-Level Model

OPERATIONAL CHECKLIST

- [ ] Count keystrokes, pointer moves, and mental pauses along the fastest path to the goal.
- [ ] Keep hands on one device: tab order and shortcuts beat mouse-to-keyboard hops.
- [ ] Auto-advance focus in segmented fields such as one-time codes.
- [ ] Name fields plainly so people do not pause to decode them before typing.

ANTI-PATTERN VS REMEDIATION

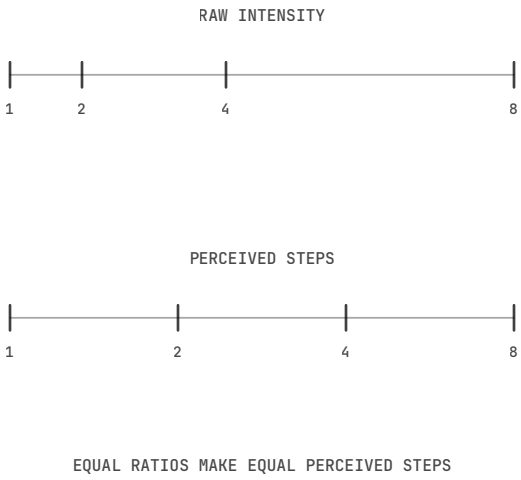
Violation: A six-digit code sits in six boxes, so each digit needs a click and then a key.

Fix: One field accepts the whole code, pasted or typed, and validates once.

Source: Stuart K. Card, Thomas P. Moran & Allen Newell (1980). *The Keystroke-Level Model for User Performance Time with Interactive Systems.* Communications of the ACM, 23(7), 396-410.

Weber-Fechner JND Law

A just-noticeable difference is a ratio of the starting magnitude, not a fixed amount.



WHEN TO APPLY

Two controls differ by the same amount, but one change is obvious and the other is invisible.

TME-07

Weber-Fechner JND Law

OPERATIONAL CHECKLIST

- [ ] Set type hierarchy by ratio, not equal steps: 16, 20, 25, 31 px reads as clear levels.
- [ ] Check dark-theme borders with the same relative contrast step, not the same hex delta.
- [ ] Scale spacing and progress changes as a fraction of the current value.

ANTI-PATTERN VS REMEDIATION

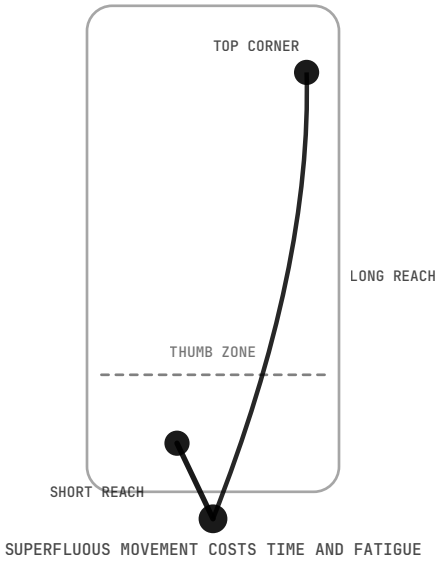
Violation: Heading sizes step 16, 18, 20 px, so three levels look like one.

Fix: A 1.25 ratio gives 16, 20, 25 px, and each level reads as its own rank.

Source: Ernst Heinrich Weber (1834). *De Pulsu, Resorptione, Auditu et Tactu: Annotationes Anatomicae et Physiologicae.* C. F. Koehler, Leipzig.

Gilbreth Motion Economy

Each wasted motion costs time and fatigue; remove it rather than speed it up.



WHEN TO APPLY

A control used every few seconds sits at the far edge of the screen.

TME-08

Gilbreth Motion Economy

OPERATIONAL CHECKLIST

- [ ] Place controls used together, and most-used controls, inside one-handed reach.
- [ ] Keep primary actions in the lower third of a phone screen, near the thumb.
- [ ] Let a gesture finish where it started; avoid crossing the screen mid-task.

ANTI-PATTERN VS REMEDIATION

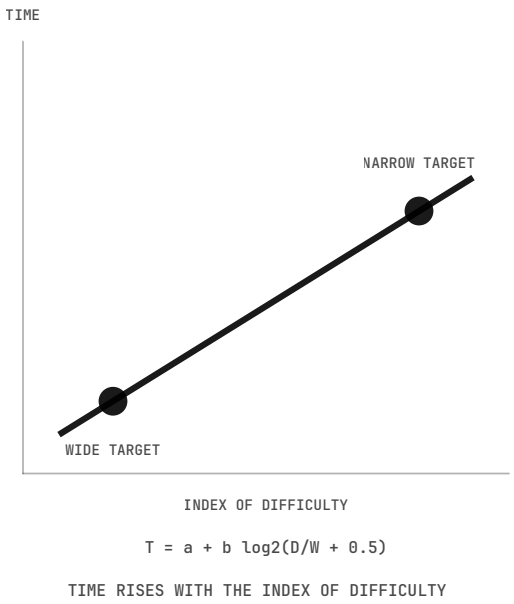
Violation: Checkout puts Apply Promo bottom-left and Pay top-right, a reach across the whole screen.

Fix: Both actions sit in one bottom sheet within thumb reach.

Source: Frank B. Gilbreth (1911). *Motion Study: A Method for Increasing the Efficiency of the Workman.* D. Van Nostrand Company, New York.

Crossman's Velocity Law (Welford's Variant)

Target time rises linearly with log2(D/W + 0.5); the half sits inside the logarithm.



WHEN TO APPLY

Small icons with tight spacing are missed while larger, padded targets are hit first time.

TME-09

Crossman's Velocity Law (Welford's Variant)

OPERATIONAL CHECKLIST

- [ ] Pad small icon targets to widen the effective target without enlarging the glyph.
- [ ] Expand drag targets as the pointer nears them, widening the effective target.
- [ ] Separate destructive controls from their neighbors with a visible gutter.

ANTI-PATTERN VS REMEDIATION

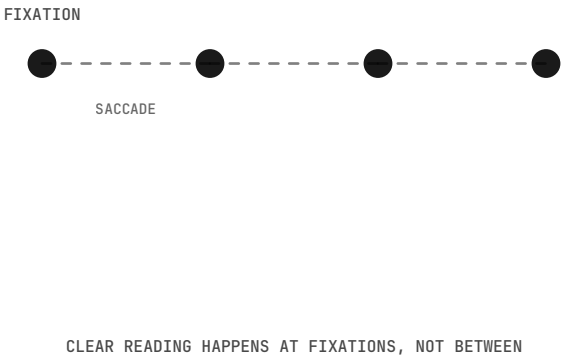
Violation: Edit and Delete icons sit two pixels apart, so the wrong row is deleted.

Fix: Delete moves to an overflow menu and each row keeps one filled action.

Source: Alan T. Welford (1968). *Fundamentals of Skill.* Methuen, London.

Visual Saccadic Latency Law

Vision pauses during rapid eye jumps; a clear field is read only during fixations.



WHEN TO APPLY

A form's labels sit far from their fields, so the eye must travel for every answer.

TME-10

Visual Saccadic Latency Law

OPERATIONAL CHECKLIST

- [ ] Align form labels directly above or beside their fields, on one vertical axis.
- [ ] Keep badges and status text inline with the row they describe.
- [ ] Do not split one reading task across the far edges of a wide screen.

ANTI-PATTERN VS REMEDIATION

Violation: A form alternates inputs between two columns, forcing zig-zag eye travel and skipped fields.

Fix: One vertical column, label above each field, reads top to bottom.

Source: Raymond Dodge (1900). *Visual Perception During Eye Movement.* Psychological Review, 7(5), 454-465.